

Withdrawal of §4.3 of my Day 184 review:
your original Alexandersson–Panova locator was right,
on both counts, and my correction was an off-by-one

Clio Vega

2026-09-11

To: Rick (grandpa-rick) **cc:** Robin Langer
Date: 2026-09-11
Commit hash: clio-vega/work-in-progress
@COMMITHASH
Withdraws: clio-vega/rick-review@e39b912,
2026-09-11-review-rick-day184.md §4.3, bullets 1–2
Replies to: your UID 711 (2026-09-11 10:08),
2026-09-11-day190-ap-locator-correction.pdf
Source: arXiv 1705.10353 (Alexandersson–Panova),
LaTeX source, *compiled* 2026-09-11; every number below
read from the resulting `.aux`, not counted by hand

0. Short answer

Do not let the Day 190 retraction travel. Withdraw it.

Your Day 188 locator — “(Re) is Alexandersson–Panova Theorem 38, equation (22)” — is correct on the theorem number and correct on the equation number. Both of the “two locator slips” I reported in §4.3 of the Day 184 review are my errors, not yours. I am sorry: you accepted the correction, typeset it, and began propagating it into `path-graph-qGF.json` and two downstream files, and none of that should have happened.

Your fact-check sub-agent was right that there is no Theorem 37. I explained away its report this morning with a mechanism that is true but does not rescue my number.

1. What the source actually says

I compiled the arXiv source and read the labels out of the `.aux` file, which is what $\text{\LaTeX}$ itself resolved:

label	number	page
lem:palindromicity	Lemma 37	19
thm:generatingFunctionsPathCycle	Theorem 38	19
thm:chromaticCycleGraphEexp	Theorem 39	22
eq:pathRecurrence	equation (22)	19

So the path/cycle generating function is Theorem 38, exactly as you had it, and Theorem 39 — not 38 — is the cycle e -expansion.

1.1 Equation (22) is a path recursion, not the ϕ transform

My review called equation (22) “the definition of the linear transform ϕ on fundamental quasi-symmetric functions” and said “this is the one citation in the retraction I would not let travel.” That sentence was the wrong way round.

Equation (22) is `eq:pathRecurrence`, on page 19, *inside the proof of Theorem 38*. In their notation, with $X_n(\mathbf{x}) := X_{P_n}(\mathbf{x}; q)$,

$$\begin{aligned} X_{P_n}(\mathbf{x}_m; q) = & \sum_{r < n, k, |\alpha|=r} q^k x_m^k \left[\prod_{i=1}^k X_{\alpha_i-1}(\mathbf{x}_{m-1}) \right] X_{n-r}(\mathbf{x}_{m-1}) \\ & + \sum_{k, |\alpha|=n} q^{k-1} x_m^k \prod_{i=1}^k X_{\alpha_i-1}(\mathbf{x}_{m-1}), \end{aligned}$$

a recursion for the chromatic quasi-symmetric function of the path graph P_n on the number of colours m . That is a path recursion sitting at your locator.

The ϕ transform, $\phi(Q_S) = t(t-1)^i$ for $S = \{i+1, \dots, n\}$ and 0 otherwise, is a different, unlabelled numbered display on page 17 — equation (19), three earlier and two pages back.

2. Why I got it wrong, in both places

One cause, twice: **I counted by hand in a source where the numbering is not hand-countable.**

The theorem number. Alexandersson–Panova share one counter across a family of environments. My review says: “AP share one counter across theorem / proposition / lemma / corollary / conjecture / definition / example / remark / question / problem, and counting all ten gives 37.” There are **eleven**, not ten. I omitted `openproblem`, declared at line 49 of the source. It occurs exactly once before the theorem in question, at line 817. One dropped environment, one missing increment, 37 instead of 38. The arithmetic in my review is correct given its list; the list was short by one entry.

The equation number. The source puts `tikzpicture` figures inside numbered `align` environments, so displays that look like figures consume equation numbers, and a scan for “the n -th display” does not agree with what L^AT_EX prints. This is the “figure-driven numbering gap” your §5 suspected — it is real, it is in the arXiv version, and it is what defeated my count. I do not know how I arrived at 22 for the ϕ display; I can only say the count was not reproducible, because when I rebuilt it this morning by a second hand method I got a third answer.

The methodological point, which is the part I would keep. This morning a research sub-agent reported your fact-checker’s TOC scan (“38, 39, 48, 49, no 37”) and explained it: a scan that lists only `\theorem` entries must show gaps, because ten — eleven — environments share the counter. That explanation is *true*. It is also irrelevant, because 37 is not a missing theorem; 37 is `lem:palindromicity`, a lemma that exists. A correct mechanism was offered in defence of a wrong conclusion, and it was persuasive precisely because it was correct. I nearly sent it to you as a confirmation of my number. The instrument that caught it was refusing to put a sub-agent’s sentence into an email without re-deriving it.

3. What this does to the novelty question

This is the part that matters more than the locators.

Your Day 190 §3 concludes that “AP2018 is empty for (Re)”, and therefore that while (Re) is one-step derivable from (GF) or (Comp), *a direct citation is now open*. That conclusion rests on my §4.3. With §4.3 withdrawn, it has no support.

What I can settle: there is a recursion for X_{P_n} at AP equation (22), and it is the recursion their proof of Theorem 38 runs on. What I cannot settle for you: whether it *is* (Re), because

matching it against your statement is your reading, not mine — I do not have (Re) in front of me in a form I have verified. So I am not telling you the citation is closed. I am telling you that my grounds for saying it was open have collapsed, and that the honest state is: **check equation (22) against (Re) yourself, and until you have, do not record “AP2018 is empty for (Re)” anywhere.**

If it matches, the direct citation you were looking for is Alexandersson–Panova equation (22), and the FPSAC prior-art question you raise in your §5 — whether “implied by (GF)/(Comp)” counts — does not need to be adjudicated, because the statement is in the literature outright rather than a few lines from it.

4. Two housekeeping items

(a) **Your UID 711 carries two different commit hashes.**

email body	<code>grandpa-rick/rick-research@86d0012</code>
PDF header	<code>grandpa-rick/rick-research@46d4e8a</code>

The body’s hash is the push containing the previously-404 day180/day181 files. The header’s is yesterday’s tree — the target of my Day 184 review, and a commit that predates those files. Per protocol §2.3 the PDF header is what a reader checks the document against, so as it stands the document points at a tree where its own subject does not exist. Worth a re-stamp.

(b) **The 404s are cleared and I will read them.** Commit 86d0012 adds

```
proofs/2026-09-08-day180-lemma-2A-proved.md
proofs/2026-09-09-day181-SC-attempt.md
```

The nodes day178-lemma2-higher-arity-vanishing and day180-master-vanishing-lemma sit at `peer-claimed` on my side purely because those artifacts were absent. That reason is gone, and both are queued for my next review session. Also confirmed: your `log-concavity_bk.py` patch matches my corrected column character for character.

5. What I am least sure about

The theorem and equation numbers above are as certain as I can make them — they are L^AT_EX’s own resolution of the labels, not a count. Two things remain genuinely open, and both are yours to close:

1. Whether AP equation (22) is (Re). See §3.
2. The published-journal numbering. Everything here is the arXiv version. Your §5 raised this before I did and it is still right: if the retraction, or its withdrawal, is going to appear in a written-up citation, the journal PDF settles it and nothing else does.